

MACHI RAJU HARI RAM

LinkedIn: [HARIRAM MACHIRAJU | LinkedIn](#)

E- mail: machirajuhariramm@gmail.com

Phone: +91 7995307237

CARRER OBJECTIVES

To pursue a highly rewarding career, seeking for a job in challenging and healthy work environment where I can utilize my innovative skills and knowledge efficiently for growth of the organization and myself as well as in this innovative and competitive world.

CARRER SUMMARY

- Debugged input errors and ensured accurate password detection.
- Familiar with communication protocols like UART, SPI, I2C, MODBUS.
- Hands-on experience with Arduino and 8051 microcontrollers through academic projects.
- Proficient in using development tools like Arduino IDE, Keil, Eclipse, Proteus and QXDM.
- Good understanding of embedded system concepts like GPIO, Timers, Interrupts and Bootloader.
- Quick learner with the ability to adapt to new technologies.
- Strong debugging and problem-solving skills developed through project-based learning

SKILLS

- ❖ **Languages:** C, Embedded C, C++, Python, Shell Scripting.
- ❖ **Communication Protocols:** CAN, SPI, I2C, UART, Ethernet, MODBUS.
- ❖ **Software Tools & IDEs:** Eclipse, Arduino IDE, Keil μ Vision, Proteus, STM32CubeIDE, QXDM.
- ❖ **Debugging Tools:** GDB, JTAG, Logic Analyzer, Oscilloscope.
- ❖ **Microcontroller & Boards:** Arduino, 8051, STM32, Raspberry Pi.
- ❖ **Concepts:** GPIO, Timers, Interrupts, Power Optimization, Memory Management, Bootloader.
- ❖ **Operating Systems:** Window, Linux (Ubuntu), Basic Linux Terminal & Shell Scripting.
- ❖ **Scripting & Automation:** Python for automation, Logging and testing; Bash for build scripts.

PROJECTS

1. SMART DOOR LOCK SYSTEM USING ARDUINO

Description: Designed a basic smart door locking mechanism using an Arduino microcontroller and keypad. The system unlocks the door only when the correct password is entered.

- Interfaced the keypad and servo motor with Arduino.
- Wrote Embedded C code to authenticate input and trigger door unlock.
- Simulated the setup using Proteus and tested with real hardware.
- Debugged input errors and ensured accurate password detection.

2. SMART TRAFFIC MANAGEMENT SYSTEM USING ARDUINO & IoT

Description: Developed an intelligent traffic control system using sensors and microcontrollers to optimize traffic flow and prioritize emergency vehicles using real-time data and wireless communication.

- Interfaced IR sensors and Arduino to monitor vehicle density at intersections.
- Designed adaptive traffic light control logic based on sensor input using Embedded C.

- Integrated RF modules for emergency vehicle detection and priority signaling.
- Simulated and tested the system in Proteus and deployed on real-time Arduino setup.
- Configured data upload to Thing Speak cloud for live monitoring and analytics.

EDUCATION

Rajeev Gandhi Memorial college of Engineering and Technology

A.P, India

Bachelor of Technology (E.C.E). GPA: 7.42

December 2021 – April 2025

Sri Gayatri Junior College

A.P, India

Intermediate (M.P.C), Marks: 904

June 2019 – July 2021

CERTIFICATIONS

1. Object-Oriented Programming with Python (Codio)

Link: [Codio python3.1 Certificate | edX](#)

May 2024

- Mastered fundamentals python syntax, proficiently utilizing control flow, loops, functions, data structures.
- Acquired expertise in procedural programming paradigms and associated logical concepts, enhancing capabilities.

2. IoT System Design: Software and Hardware Integration (Waseda University).

Link: [WasedaX IoT111x Certificate | edX](#)

April 2024

- Mastered hands-on experience in designing end-to-end IoT systems by integrating sensors, microcontrollers, and cloud platforms for real-time data monitoring and control.

3. Big Data, Hadoop, Spark Basics (IBM)

Link: [IBM BD0225EN Certificate | edX](#)

Jan 2024

- Learned about the data ecosystem, including the ETL process and big data basics.
 - Mastered data gathering, identification, and clearing for analysis preparation.
-

ACHEIVEMENTS

- IEEE YESISIT12 2024 Preliminary Round Participant by RGM CET, A.P.
- Smart India Hackathon Participant.
- Chief Information Officer (CIO) by Student Activity Club (SAC) || RGM CET.